



New Classes of Stellar Dynamics

Mathematicians from RDUN University develop new classes of stellar dynamics systems. <http://digit.in/mar21-27>



Solving Telegraph Equations

Mathematician suggests a way for solving problems of elliptic-telegraph and differential equations. <http://digit.in/mar21-28>

Credit: Martin Krzywinski



The Fundamental Constants

constants. It is a brilliant example of machines outperforming humans in the world of mathematics. What makes it particularly interesting is the way it works, the algorithms involved in its inner workings and the implications it has for modern mathematics.

HOW IT WORKS

Machine learning, that involves algorithms to detect patterns in large amounts of data, already has multiple existing applications such as image recognition. The Ramanujan Machine applies machine learning on a basic foundational concept: Numbers. Although this is not the first-time machine learning has been used in such a scenario – previously ML has been used for automated theorem proving – the implementation of the Ramanujan Machine is very different.

The Ramanujan Machine consists of two algorithms: A Meet-In-The-Middle (MITM) algorithm and a Gradient Descent (GD) which is designed for the structure of continued fractions. They match numerical values that are input to the system to a formula, thus conjecturing formulas without providing proofs, and without any prerequisite knowledge of any mathematical structures. The way it works is, a fixed numerical value (such as a fundamental constant) is input into the system, and the algorithms work to construct a continued fraction that represents the value that was inputted. It iterates through numerous equations to indicate the existence of a formula to express the input value constant. The MITM algorithm and the Gradient Descent (GD) are both based on matching numerical values,

to learn the relations between fundamental constants. While it is possible to use this method for many forms of mathematical structures, the Ramanujan Machine focuses on its potential with equations involving polynomial continued fractions (PCFs) such as the one shown below.

$$x = a_0 + \frac{b_1}{a_1 + \frac{b_2}{a_2 + \frac{b_3}{\dots}}}$$

Credit: arXiv:1907.00205 [cs.LG]

This algorithm has already produced several results that have been proven, such as this:

$$\frac{4}{3\pi - 8} = 3 - \frac{1 \cdot 1}{6 - \frac{2 \cdot 3}{9 - \frac{3 \cdot 5}{12 - \frac{4 \cdot 7}{\dots}}}}$$

Credit: arXiv:1907.00205 [cs.LG]

One may wonder whether these are just mathematical coincidences, or if they are truly mathematical identities. Do these identities break down once enough digits have been calculated? The method employed makes it discerningly unlikely for the conjectures to break down. The probability of finding an error in 10,000,000,000 digits of the calculation is lesser than 10-40. The algorithms have tested conjectures for up to 2000 digits of accuracy, however such accuracy still demands a rigorous proof for these theorems.

ITS IMPACT

Many known mathematical identities and theorems, Regular Formulas (RFs), were discovered using sequential logical steps derived from pre-existing knowledge of the properties of these

fundamental constants. Since the Ramanujan Machine reverses this process, and uses just the numerical values of the constants, every conjecture from the machine enables reverse-engineering of the mathematical structure. In some cases, the proofs use new techniques and provide new valuable insight on the field. The success of this approach, to finding new theorems for fundamental constants, opens additional avenues for more efficient algorithms and further research in the field.

After the first appearance of the work presented by the researchers of the Ramanujan Machine, many people ran the algorithms and found new conjectures. Others responded with formal proofs to the new RFs discovered by the algorithms, and in fact, over a span of a few months, proofs for all the RFs in the original published research paper were presented. These conjectures have also enabled mathematicians to make further conjectures based on the conjectures provided by the Ramanujan Machine. The researchers believe that this back-and-forth between the algorithms and mathematicians is the crux of what can be achieved through automated conjecture generation of fundamental constants.

The idea of the Ramanujan Machine is to involve the global mathematics community to come together to prove the conjectures that the algorithms suggest. However, a part of the objective is to also involve more people without a mathematics background to the process of formally proving conjectures. So far over 100 conjectures have surfaced from the machine. There are new conjectures for many mathematical constants such as Apéry's constant, Catalan's constant, e , π and the golden ratio ϕ .

The researchers of the Ramanujan Machine allow the general community to donate computational time to find more such conjectures and even allow you to name the conjecture after yourself! This seems like a novel way to go about proving conjectures, and is likely to become the future of mathematics.